Market Basket Analysis with Python Project - eBay

In [1]:
```
## import libraries :
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import scipy.stats as stats

import warnings
warnings.filterwarnings("ignore")
```

Task 1: Data Cleaning and Preparation

- Remove duplicate or inconsistent survey responses.
- Standardize categorical/numerical entries (e.g., frequency levels, gender, recommendation responses).
- Handle missing values and inconsistent formats in Product_Search_Method and otherfields.
- Rename duplicate or misformatted columns (e.g., remove trailing spaces inRating_Accuracy).
- Convert numerical rating columns (e.g., Customer_Reviews_Importance,Shopping_Satisfaction) to appropriate numeric types for analysis.

In [2]:
```
# Task 1: Data Cleaning and Preparation

## step 1.1: Load Data:

df= pd.read_csv("eBay.csv")
df.head()
```

Out[2]:

	Timestamp	age	Gender	Purchase_Frequency	Purchase_Categories	Personalized_Recommendation_Frequency	Browsing_Frequency	Product_Search_Method	Search_Result_Exploration	Custom
0	2023/06/07 11:44:55 AM GMT+5:30	32	Prefer not to say	Multiple times a week	Groceries and Gourmet Food;Home and Kitchen	Sometimes	Rarely	categories	Multiple pages	
1	2023/06/07 9:28:09 AM GMT+5:30	47	Female	Multiple times a week	Groceries and Gourmet Food;Beauty and Personal...	No	Few times a month	NaN	First page	
2	2023/06/05 10:09:03 PM GMT+5:30	50	Female	Once a month	Groceries and Gourmet Food;Beauty and Personal...	Sometimes	Multiple times a day	categories	Multiple pages	
3	2023/06/07 5:58:12 PM GMT+5:30	6	Others	Once a month	Groceries and Gourmet Food;Beauty and Personal...	No	Rarely	others	Multiple pages	
4	2023/06/07 11:46:52 AM GMT+5:30	61	Male	Once a week	Groceries and Gourmet Food;Clothing and Fashion	Yes	Rarely	NaN	Multiple pages	

5 rows × 24 columns

In [3]:
```
## check missing values and  shape of the data :

print("Data Shape:", df.shape)
print("-" * 50)
df.info()
```

```
Data Shape: (800, 24)
--------------------------------------------------
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 800 entries, 0 to 799
Data columns (total 24 columns):
 #   Column                                  Non-Null Count  Dtype
---  ------                                  --------------  -----
 0   Timestamp                               800 non-null    object
 1   age                                     800 non-null    int64
 2   Gender                                  800 non-null    object
 3   Purchase_Frequency                      800 non-null    object
 4   Purchase_Categories                     800 non-null    object
 5   Personalized_Recommendation_Frequency   800 non-null    object
 6   Browsing_Frequency                      800 non-null    object
 7   Product_Search_Method                   639 non-null    object
 8   Search_Result_Exploration               800 non-null    object
 9   Customer_Reviews_Importance             800 non-null    int64
 10  Add_to_Cart_Browsing                    800 non-null    object
 11  Cart_Completion_Frequency               800 non-null    object
 12  Cart_Abandonment_Factors                800 non-null    object
 13  Saveforlater_Frequency                  800 non-null    object
 14  Review_Left                             800 non-null    object
 15  Review_Reliability                      800 non-null    object
 16  Review_Helpfulness                      800 non-null    object
 17  Personalized_Recommendation_Frequency   800 non-null    int64
 18  Recommendation_Helpfulness              800 non-null    object
 19  Rating_Accuracy                         800 non-null    int64
 20  Shopping_Satisfaction                   800 non-null    int64
 21  Service_Appreciation                    800 non-null    object
 22  Improvement_Areas                       800 non-null    object
 23  transaction                             800 non-null    int64
dtypes: int64(6), object(18)
memory usage: 150.1+ KB
```

In [4]:
```
## step 1.2: Remove duplicates

df=df.drop_duplicates()
```

In [5]:
```
## step 1.3: Clean Column Names

df.columns=df.columns.str.strip().str.replace(" ","_")
df.columns
```

Out[5]:
```
Index(['Timestamp', 'age', 'Gender', 'Purchase_Frequency',
       'Purchase_Categories', 'Personalized_Recommendation_Frequency',
       'Browsing_Frequency', 'Product_Search_Method',
       'Search_Result_Exploration', 'Customer_Reviews_Importance',
       'Add_to_Cart_Browsing', 'Cart_Completion_Frequency',
       'Cart_Abandonment_Factors', 'Saveforlater_Frequency', 'Review_Left',
       'Review_Reliability', 'Review_Helpfulness',
       'Personalized_Recommendation_Frequency', 'Recommendation_Helpfulness',
       'Rating_Accuracy', 'Shopping_Satisfaction', 'Service_Appreciation',
       'Improvement_Areas', 'transaction'],
      dtype='object')
```

In [6]:
```
## step 1.4: Standardize categorical

df['Gender'] = df['Gender'].str.lower()
```

```
df["Gender"]=df["Gender"].replace({
    "m":"Male","male":"Male","f":"Female","female":"Female"
})
df.head()
```

Out[6]:

	Timestamp	age	Gender	Purchase_Frequency	Purchase_Categories	Personalized_Recommendation_Frequency	Browsing_Frequency	Product_Search_Method	Search_Result_Exploration	Custom
0	2023/06/07 11:44:55 AM GMT+5:30	32	prefer not to say	Multiple times a week	Groceries and Gourmet Food;Home and Kitchen	Sometimes	Rarely	categories	Multiple pages	
1	2023/06/07 9:28:09 AM GMT+5:30	47	Female	Multiple times a week	Groceries and Gourmet Food;Beauty and Personal...	No	Few times a month	NaN	First page	
2	2023/06/05 10:09:03 PM GMT+5:30	50	Female	Once a month	Groceries and Gourmet Food;Beauty and Personal...	Sometimes	Multiple times a day	categories	Multiple pages	
3	2023/06/07 5:58:12 PM GMT+5:30	6	others	Once a month	Groceries and Gourmet Food;Beauty and Personal...	No	Rarely	others	Multiple pages	
4	2023/06/07 11:46:52 AM GMT+5:30	61	Male	Once a week	Groceries and Gourmet Food;Clothing and Fashion	Yes	Rarely	NaN	Multiple pages	

5 rows × 24 columns

In [7]:

```
## Purchase_Frequency

df["Purchase_Frequency"]=df["Purchase_Frequency"].str.capitalize()

## Recommandation response

df["Recommendation_Helpfulness"]=df["Recommendation_Helpfulness"].str.capitalize()
```

In [8]:

```
##  step 1.5: Handles missing values

df.isnull().sum()
```

Out[8]:

```
Timestamp                                 0
age                                       0
Gender                                    0
Purchase_Frequency                        0
Purchase_Categories                       0
Personalized_Recommendation_Frequency     0
Browsing_Frequency                        0
Product_Search_Method                   161
Search_Result_Exploration                 0
Customer_Reviews_Importance               0
Add_to_Cart_Browsing                      0
Cart_Completion_Frequency                 0
Cart_Abandonment_Factors                  0
Saveforlater_Frequency                    0
Review_Left                               0
Review_Reliability                        0
Review_Helpfulness                        0
Personalized_Recommendation_Frequency     0
Recommendation_Helpfulness                0
Rating_Accuracy                           0
Shopping_Satisfaction                     0
Service_Appreciation                      0
Improvement_Areas                         0
transaction                               0
dtype: int64
```

In [9]:

```
## Numerical columns ---> Mean
num_cols=[
    "Rating_Accuracy","Customer_Reviews_Importance","Shopping_Satisfaction"
]

for col in num_cols:
    df[col]=pd.to_numeric(df[col],errors="coerce")
    df[col].fillna(df[col].mean(),inplace=True)
```

In [10]:

```
## category ----> mode

cat_col=[
    "Gender","Product_Search_Method"
]

for col in cat_col:
    df[col].fillna(df[col].mode()[0],inplace=True)

df['Product_Search_Method'].isnull().sum()
```

Out[10]:

```
np.int64(0)
```

Task 2: Descriptive Behavior Analysis

- Summarize customer demographics (age, gender distribution).
- Analyze overall purchase frequency and most popular product categories.
- Identify top browsing methods and most common cart abandonment factors.
- Calculate mean and median satisfaction, recommendation helpfulness, and rating accuracy.
- Generate summary statistics and visualizations for key behavioral variables.

In [11]:

```
## Task 2: Descriptive Behavior Analysis

## step 2.1: Summarize customer demographics (age, gender distribution).

df["Gender"].value_counts()
```

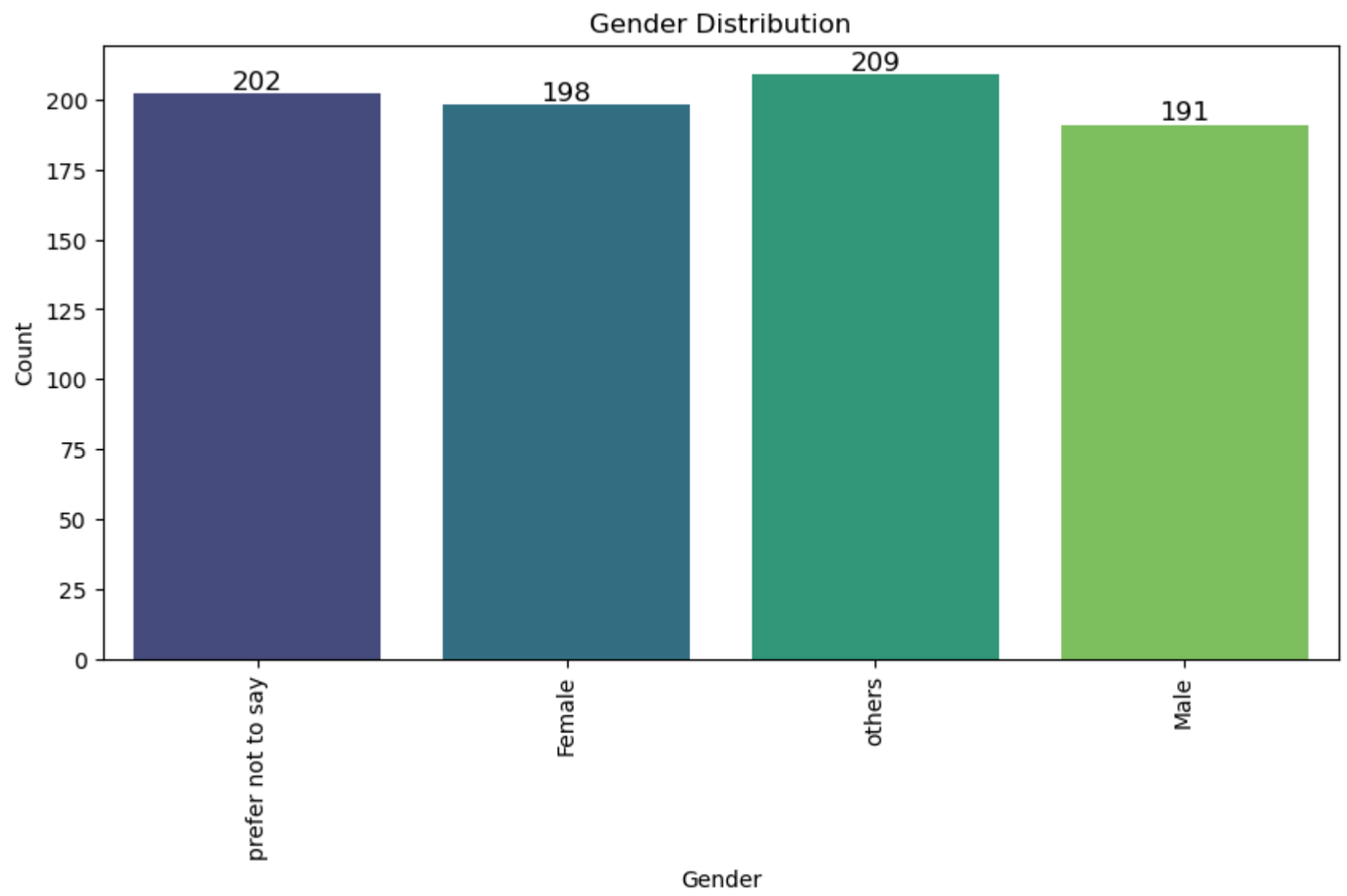
Out[11]:

```
Gender
others               209
prefer not to say    202
Female               198
Male                 191
Name: count, dtype: int64
```

In [12]:

```
plt.figure(figsize=(10, 5))

ax = sns.countplot(data=df, x="Gender", palette="viridis")

plt.xticks(rotation=90)
plt.xlabel("Gender")
plt.ylabel("Count")
```

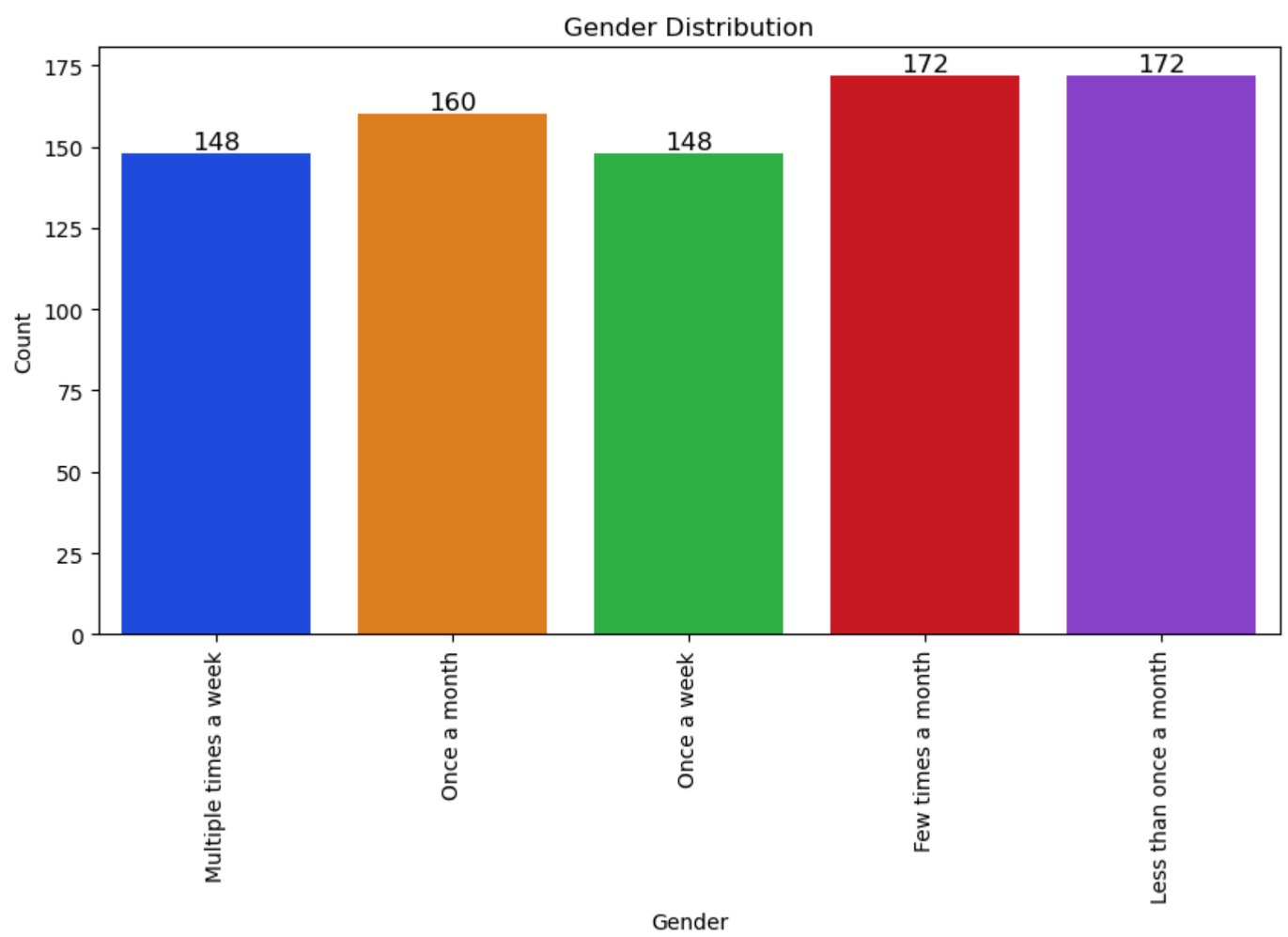
```
plt.title("Gender Distribution")

# Add data labels on bars:

for p in ax.patches:
    ax.annotate(
        f'{int(p.get_height())}',
        (p.get_x() + p.get_width() / 2., p.get_height()),
        ha='center',
        va='bottom',
        fontsize=12,
        color='Black'
    )
plt.show()
```



In [13]:
```
## step 2.2:  Analyze overall purchase frequency and most popular product categories.

df["Purchase_Frequency"].value_counts()
```

Out[13]:
```
Purchase_Frequency
Less than once a month    172
Few times a month         172
Once a month              160
Multiple times a week     148
Once a week               148
Name: count, dtype: int64
```

In [14]:
```
plt.figure(figsize=(10, 5))

ax = sns.countplot(data=df, x="Purchase_Frequency", palette="bright")

plt.xticks(rotation=90)
plt.xlabel("Gender")
plt.ylabel("Count")
plt.title("Gender Distribution")

# Add data labels on bars:

for p in ax.patches:
    ax.annotate(
        f'{int(p.get_height())}',
        (p.get_x() + p.get_width() / 2., p.get_height()),
        ha='center',
        va='bottom',
        fontsize=12,
        color='Black'
    )
plt.show()
```

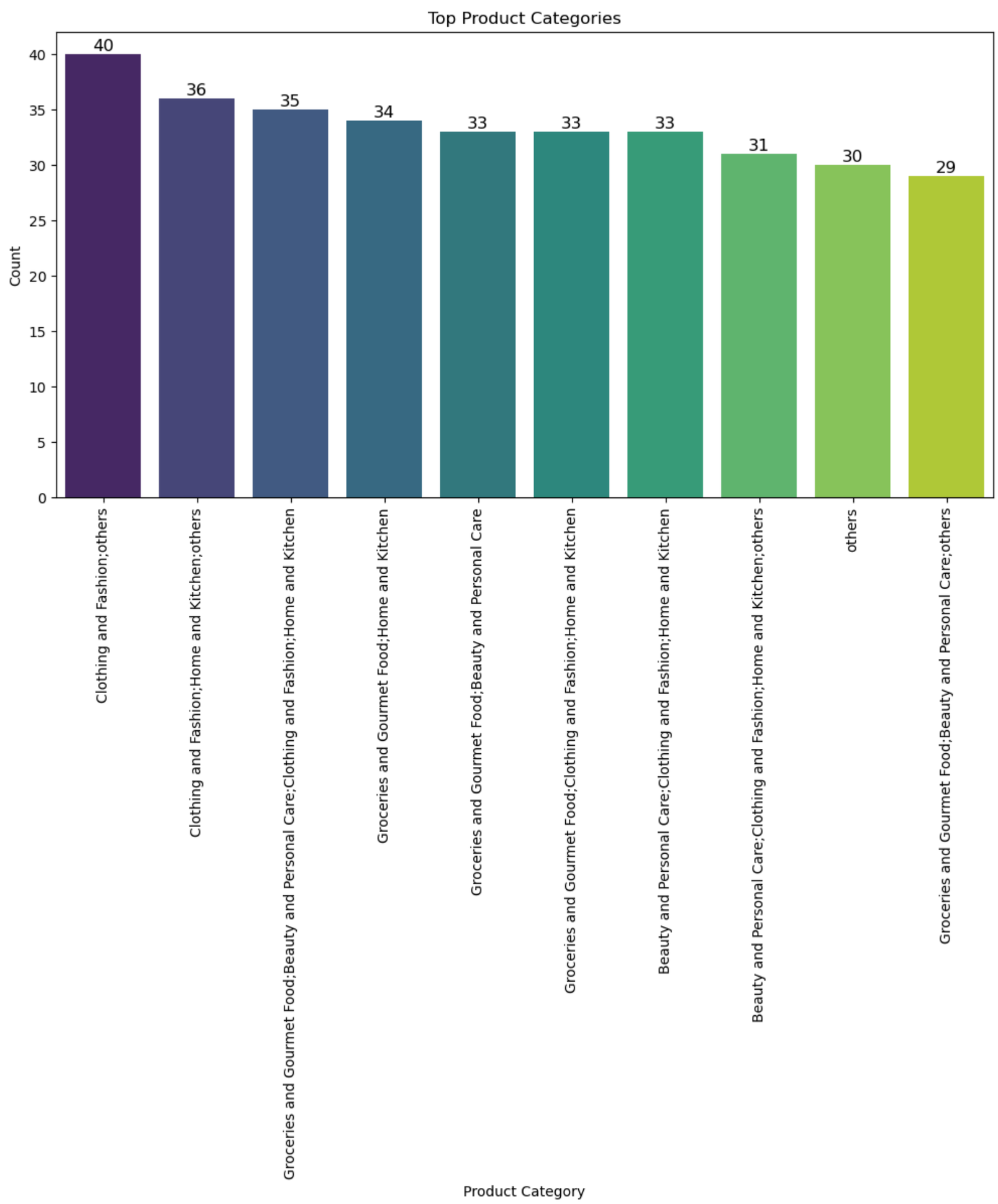


In [15]:
```
top_categories = df["Purchase_Categories"].value_counts().head(10)
print(f"Top Product Categories:\n{top_categories}")
```

```
Top Product Categories:
Purchase_Categories
Clothing and Fashion;others                                                          40
Clothing and Fashion;Home and Kitchen;others                                         36
Groceries and Gourmet Food;Beauty and Personal Care;Clothing and Fashion;Home and Kitchen    35
Groceries and Gourmet Food;Home and Kitchen                                          34
Groceries and Gourmet Food;Beauty and Personal Care                                  33
Groceries and Gourmet Food;Clothing and Fashion;Home and Kitchen                     33
Beauty and Personal Care;Clothing and Fashion;Home and Kitchen                       33
Beauty and Personal Care;Clothing and Fashion;Home and Kitchen;others                31
others                                                                               30
Groceries and Gourmet Food;Beauty and Personal Care;others                           29
Name: count, dtype: int64
```

In [16]:

```python
## create Count_plot of Top_Categories:

plt.figure(figsize=(12,6))

counts = df['Purchase_Categories'].value_counts().head(10)

ax = sns.barplot(
    x=counts.index,
    y=counts.values,
    palette='viridis'
)

plt.title("Top Product Categories")
plt.xlabel("Product Category")
plt.ylabel("Count")
plt.xticks(rotation=90)

#  Add data labels
for i, v in enumerate(counts.values):
    ax.text(
        i,
        v,
        str(v),
        ha='center',
        va='bottom',
        fontsize=12
    )
plt.show()
```

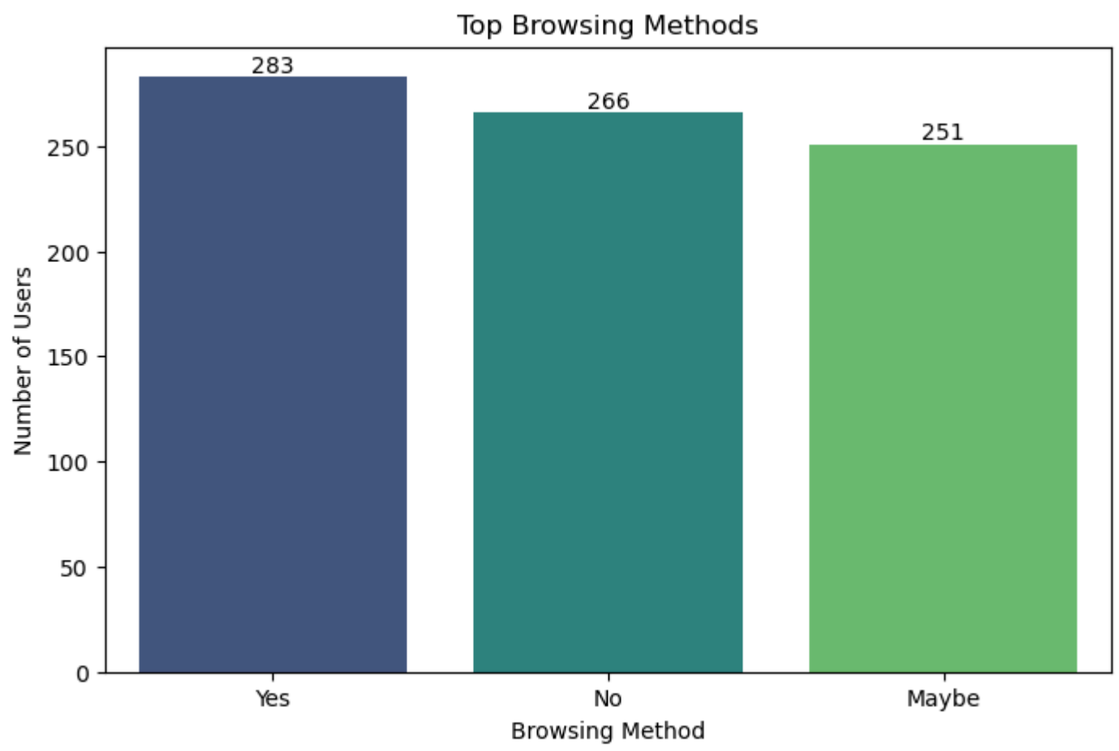


In [17]:

```python
## step 2.3: Identify top browsing methods and most common cart abandonment factors.

# a:) Top browsing methods
top_browsing = df['Add_to_Cart_Browsing'].value_counts()

print("Top Browsing Methods:")
print(top_browsing)
```

```
Top Browsing Methods:
Add_to_Cart_Browsing
Yes      283
No       266
Maybe    251
Name: count, dtype: int64
```

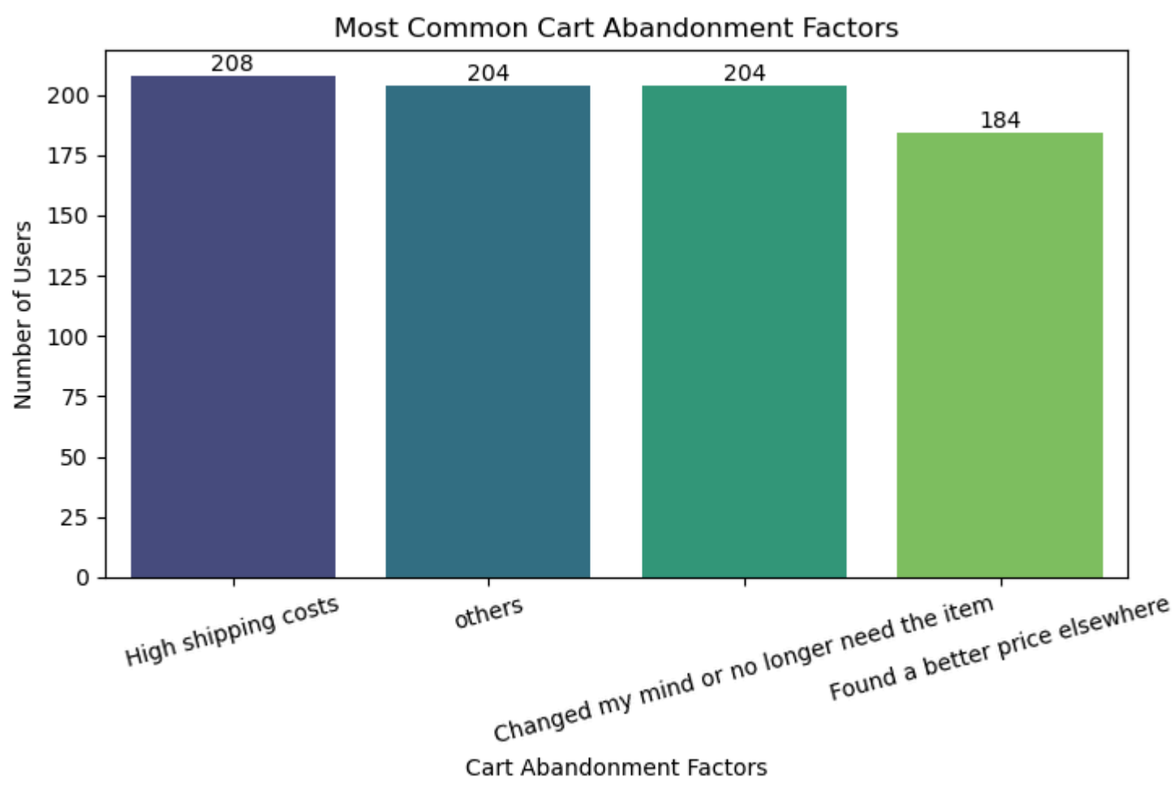
In [18]:
```
plt.figure(figsize=(8,5))

ax = sns.barplot(
    x=top_browsing.index,
    y=top_browsing.values,
    palette='viridis'
)

plt.title("Top Browsing Methods")
plt.xlabel("Browsing Method")
plt.ylabel("Number of Users")

# Data labels
for i, v in enumerate(top_browsing.values):
    ax.text(i, v, str(v), ha='center', va='bottom')

plt.show()
```



In [19]:
```
## b:) Most common Cart_Abandonment_Factors :
Cart_Abandonment_Factors = df['Cart_Abandonment_Factors'].value_counts()

print("Cart_Abandonment_Factors_Method:")
print(Cart_Abandonment_Factors)
```

```
Cart_Abandonment_Factors_Method:
Cart_Abandonment_Factors
High shipping costs                          208
others                                       204
Changed my mind or no longer need the item   204
Found a better price elsewhere               184
Name: count, dtype: int64
```

In [20]:
```
plt.figure(figsize=(7.5,5))

ax = sns.barplot(
    x=Cart_Abandonment_Factors.index,
    y=Cart_Abandonment_Factors.values,
    palette='viridis'
)

plt.title("Most Common Cart Abandonment Factors")
plt.xlabel("Cart Abandonment Factors")
plt.ylabel("Number of Users")
plt.xticks(rotation=15)

#  Data labels :
for p in ax.patches:
    ax.annotate(
        f'{int(p.get_height())}',
        (p.get_x() + p.get_width() / 2., p.get_height()),
        ha='center',
        va='bottom',
        fontsize=10
    )
plt.tight_layout()
plt.show()
```



In [21]:
```
## step 2.4: Calculate mean and median satisfaction, recommendation helpfulness, and rating accuracy.

## Mean :
df[['Shopping_Satisfaction','Rating_Accuracy']].mean()
```

```
## Median :
df[['Shopping_Satisfaction','Rating_Accuracy']].median()
```

Out[21]:
```
Shopping_Satisfaction    3.0
Rating_Accuracy          3.0
dtype: float64
```

Task 3: Customer Segmentation and Profiling.

- Segment customers based on purchase frequency and shopping satisfaction levels.
- Create profiles such as:
  - Frequent Buyers: High purchase frequency, high satisfaction.
  - Occasional Shoppers: Medium frequency, moderate satisfaction.
  - At-Risk Customers: Low satisfaction or frequent cart abandonment.
- Analyze demographic or behavioral differences across these segments.
- Use clustering (e.g., K-Means) for behavioral grouping based on survey responses.

In [22]:
```
## step 3.1: Segment customers based on purchase frequency and shopping satisfaction levels.

## Rule-based Segment:
def segment_customer(row):
    if row["Purchase_Frequency"] == "High" and row["Shopping_Satisfaction"]>=3:
        return "Frequent Buyers"
    elif row["Purchase_Frequency"]=="Medium":
        return "Occasional Shoppers"
    else:
        return "At-Risk Customers"
df["Customer_Segment"]=df.apply(segment_customer,axis=1)
```

In [23]:
```
df["Customer_Segment"].value_counts()
```

Out[23]:
```
Customer_Segment
At-Risk Customers    800
Name: count, dtype: int64
```

In [24]:
```
df["Customer_Segment"]
```

Out[24]:
```
0      At-Risk Customers
1      At-Risk Customers
2      At-Risk Customers
3      At-Risk Customers
4      At-Risk Customers
             ...
795    At-Risk Customers
796    At-Risk Customers
797    At-Risk Customers
798    At-Risk Customers
799    At-Risk Customers
Name: Customer_Segment, Length: 800, dtype: object
```

In [25]:
```
## step 3.2 : Segment Profile.

df.groupby('Customer_Segment')["Shopping_Satisfaction"].mean()
```

Out[25]:
```
Customer_Segment
At-Risk Customers    2.86625
Name: Shopping_Satisfaction, dtype: float64
```

In [28]:
```
df.groupby('Customer_Segment')['Cart_Abandonment_Factors'].value_counts()
```

Out[28]:
```
Customer_Segment   Cart_Abandonment_Factors
At-Risk Customers  High shipping costs                          208
                   Changed my mind or no longer need the item   204
                   others                                       204
                   Found a better price elsewhere               184
Name: count, dtype: int64
```

In [35]:
```
df['Customer_Segment'].value_counts()
```

Out[35]:
```
Customer_Segment
At-Risk Customers    800
Name: count, dtype: int64
```

In [39]:
```
## step 3.4: Use clustering (e.g., K-Means) for behavioral grouping based on survey responses.

# import libraries :
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
```

In [65]:
```
## K-Means Clustering

scatter_features= ['Shopping_Satisfaction',
                   'Customer_Reviews_Importance',
                   'Rating_Accuracy']
```

In [66]:
```
# Scale Data :

X=df[scatter_features]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

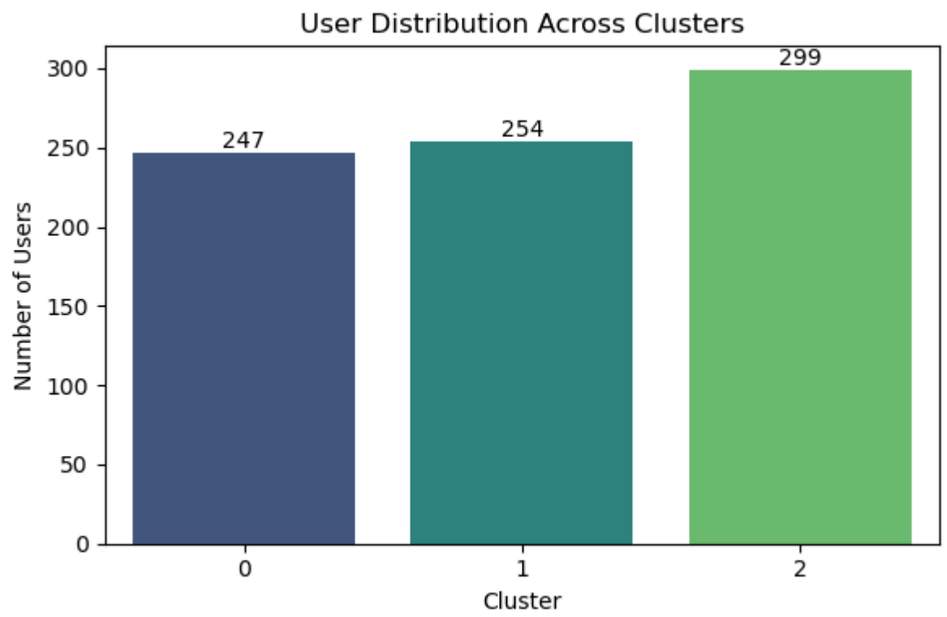
In [68]:
```
# Apply KMeans

kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
df["Cluster"] = kmeans.fit_predict(X_scaled)
```

In [69]:
```
df['Cluster'].value_counts()
```

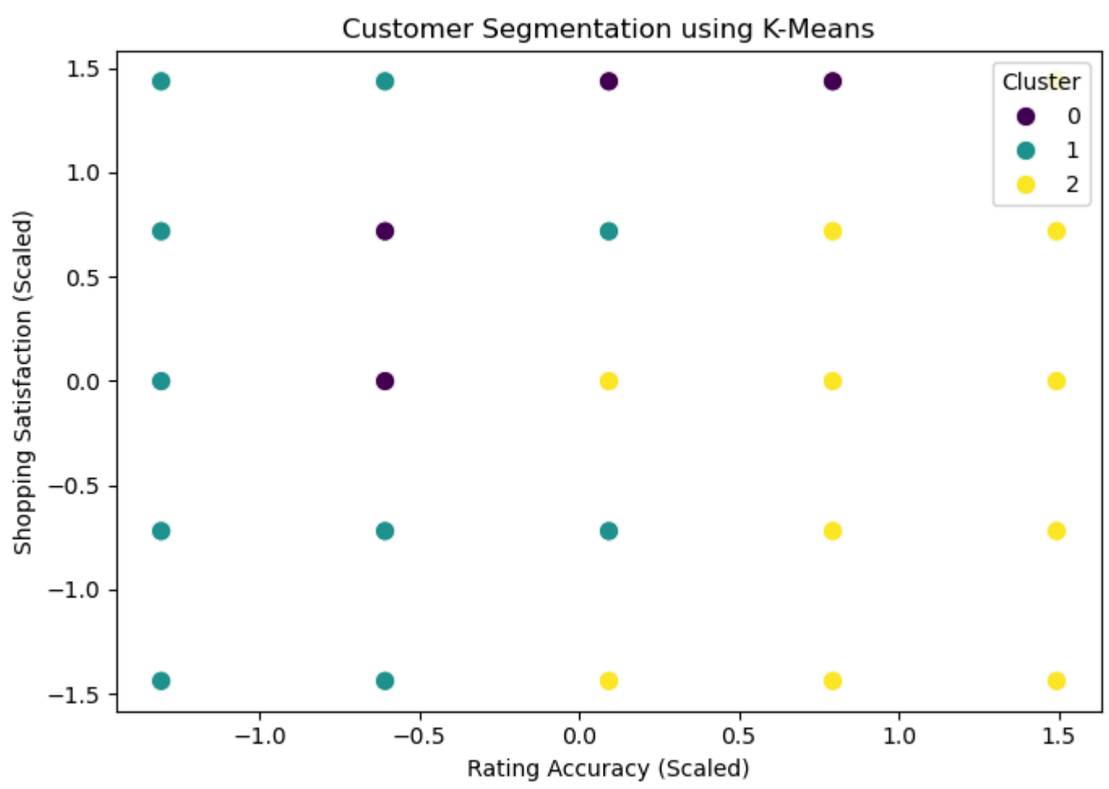
Out[69]:
```
Cluster
2    299
1    254
0    247
Name: count, dtype: int64
```

In [77]:
```
plt.figure(figsize=(6,4))

ax = sns.countplot(x='Cluster', data=df, palette='viridis')

plt.title("User Distribution Across Clusters")
plt.xlabel("Cluster")
plt.ylabel("Number of Users")

# Data labels
for p in ax.patches:
    ax.annotate(
        int(p.get_height()),
        (p.get_x() + p.get_width() / 2., p.get_height()),
        ha='center', va='bottom'
```

```
    )

plt.tight_layout()
plt.show()
```



In [79]:
```
## scatter plot :

plt.figure(figsize=(7,5))

sns.scatterplot(
    x=X_scaled[:, 0],
    y=X_scaled[:, 1],
    hue=df["Cluster"],
    palette="viridis",
    s=80
)

plt.title("Customer Segmentation using K-Means")
plt.xlabel("Rating Accuracy (Scaled)")
plt.ylabel("Shopping Satisfaction (Scaled)")
plt.legend(title="Cluster")
plt.tight_layout()
plt.show()
```



Task 4: Recommendation and Review Insights

- Examine the relationship between recommendation helpfulness and shopping satisfaction.
- Evaluate how review reliability and helpfulness impact overall ratings.
- Identify trends in how often customers engage with or trust personalized recommendations.
- Suggest actionable insights for improving eBay's recommendation system.

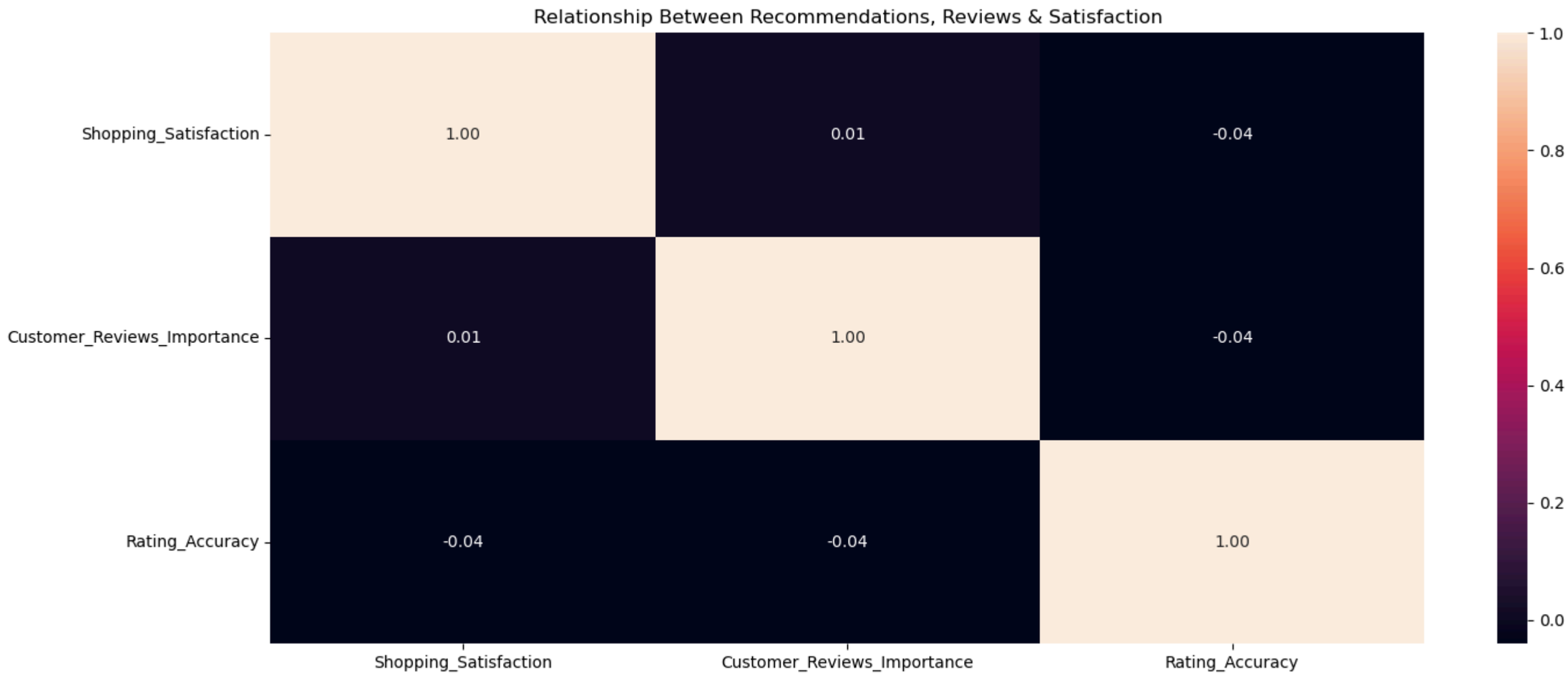
In [80]:
```
# step 4.1:  Examine the relationship between recommendation helpfulness and shopping satisfaction.
correlation=df[['Shopping_Satisfaction',
               'Customer_Reviews_Importance',
               'Rating_Accuracy']].corr()
```

In [81]:
```
correlation
```

Out[81]:

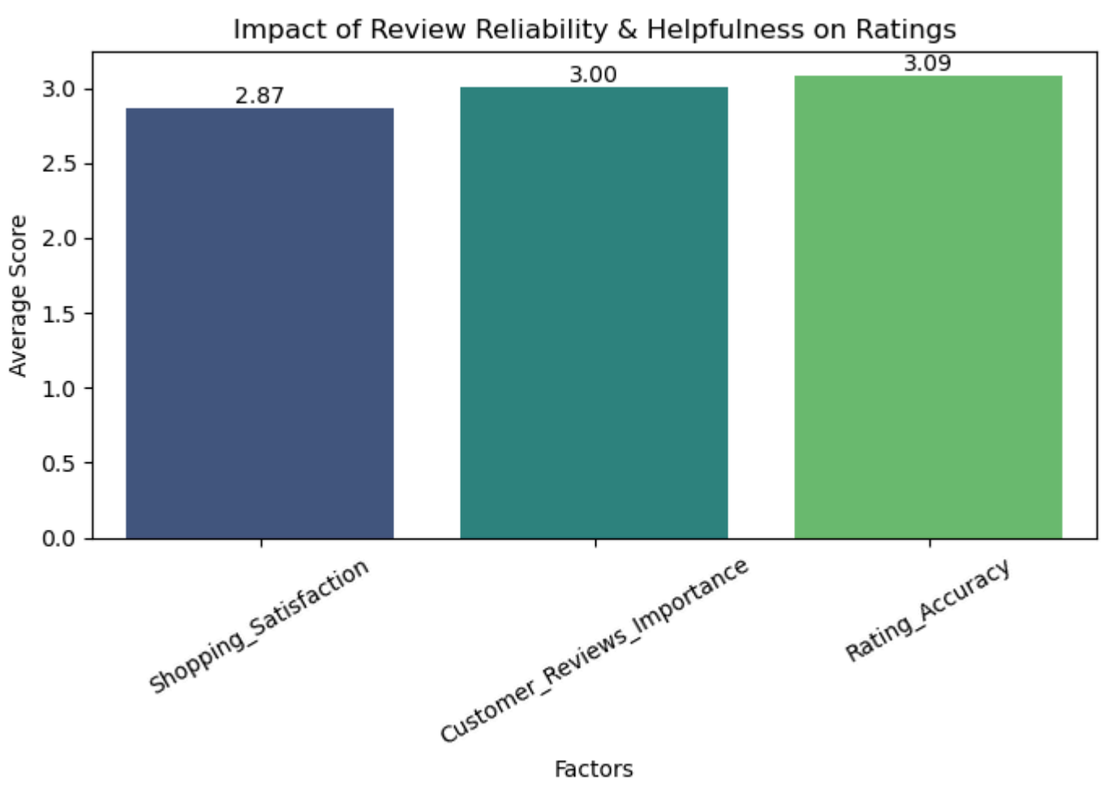
	Shopping_Satisfaction	Customer_Reviews_Importance	Rating_Accuracy
Shopping_Satisfaction	1.000000	0.012039	-0.039912
Customer_Reviews_Importance	0.012039	1.000000	-0.037404
Rating_Accuracy	-0.039912	-0.037404	1.000000

In [99]:
```
## create heatmap chart :

plt.figure(figsize=(15,6))

sns.heatmap(correlation,annot=True,fmt=".2f")
plt.title("Relationship Between Recommendations, Reviews & Satisfaction")
plt.tight_layout()
plt.show()
```



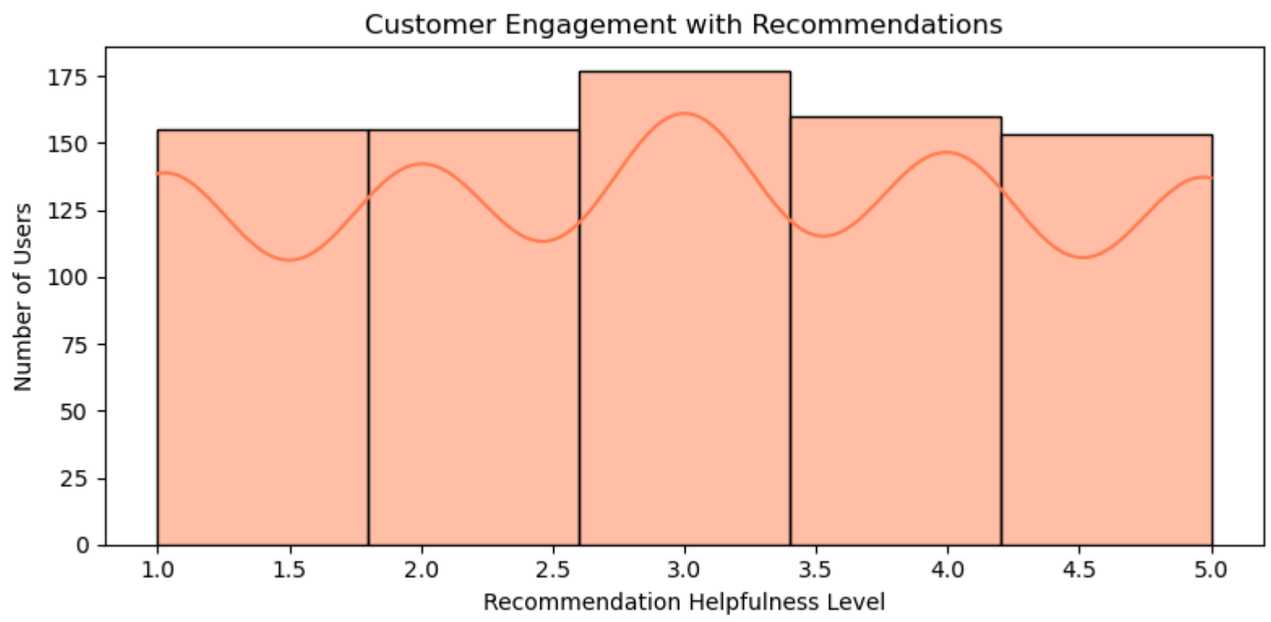
In [117…

```
# Impact of Review Reliability & Helpfulness on Ratings

mean_scores = df[
    ['Shopping_Satisfaction',
     'Customer_Reviews_Importance',
     'Rating_Accuracy']
].mean()

plt.figure(figsize=(7,5))
ax = sns.barplot(
    x=mean_scores.index,
    y=mean_scores.values,
    palette='viridis'
)

plt.title("Impact of Review Reliability & Helpfulness on Ratings")
plt.xlabel("Factors")
plt.ylabel("Average Score")
plt.xticks(rotation=30)

# Data labels
for p in ax.patches:
    ax.annotate(
        f"{p.get_height():.2f}",
        (p.get_x() + p.get_width()/2, p.get_height()),
        ha='center',
        va='bottom'
    )

plt.tight_layout()
plt.show()
```



In [116…

```
## 4.3: Engagement Trend with Recommendations
plt.figure(figsize=(8,4))
sns.histplot(
    df['Customer_Reviews_Importance'],
    bins=5,
    kde=True,
    color='coral'
)

plt.title("Customer Engagement with Recommendations")
plt.xlabel("Recommendation Helpfulness Level")
plt.ylabel("Number of Users")
plt.tight_layout()
plt.show()
```



In [115…

```
## step 4.4: Suggest actionable insights for improving eBay's recommendation system.

print("INSIGHTS:")
print("- Recommendation helpfulness has a weak correlation with satisfaction.")
print("- Review reliability alone does not strongly impact ratings.")
print("- Customers moderately engage with personalized recommendations.")
print("- Improving transparency and trust can enhance engagement.")
```

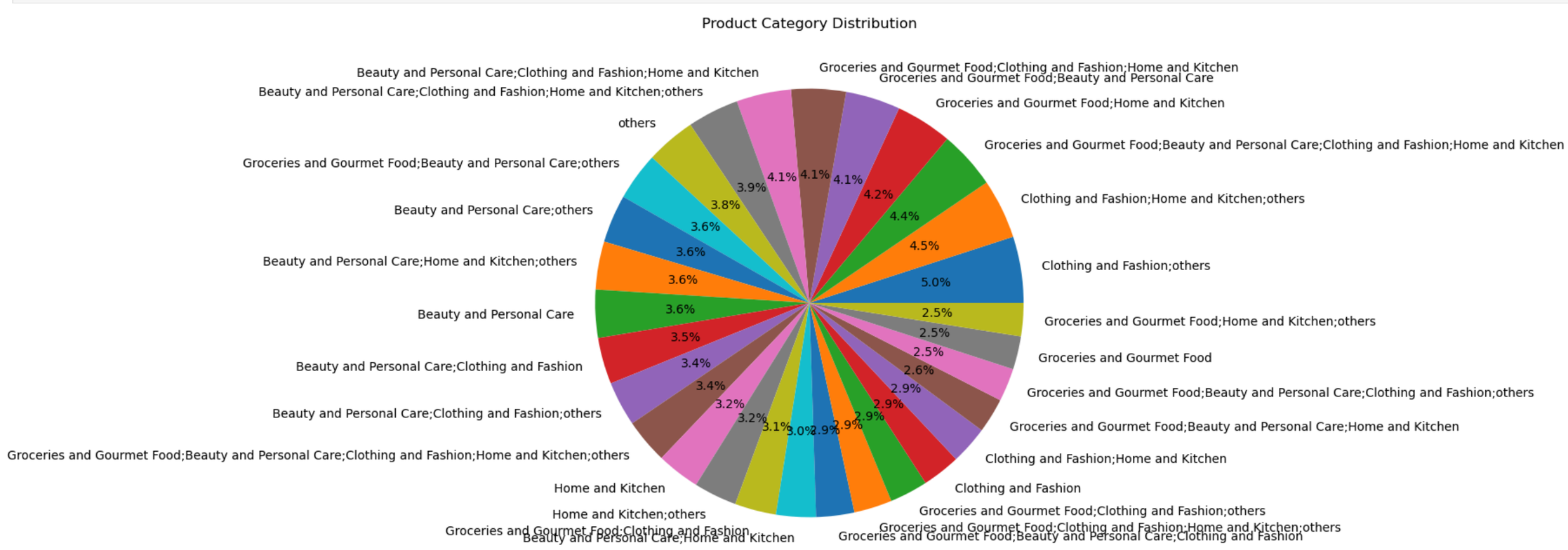
```
INSIGHTS:
- Recommendation helpfulness has a weak correlation with satisfaction.
- Review reliability alone does not strongly impact ratings.
- Customers moderately engage with personalized recommendations.
- Improving transparency and trust can enhance engagement.
```

Task 5: Visualization and Reporting.

- Create attractive visualizations (bar charts, heatmaps, pie charts) for:
  - Purchase categories
  - Browsing frequency distribution
  - Satisfaction levels
  - Correlation between recommendation usefulness and satisfaction
- Summarize findings in a clear and visually appealing report format.

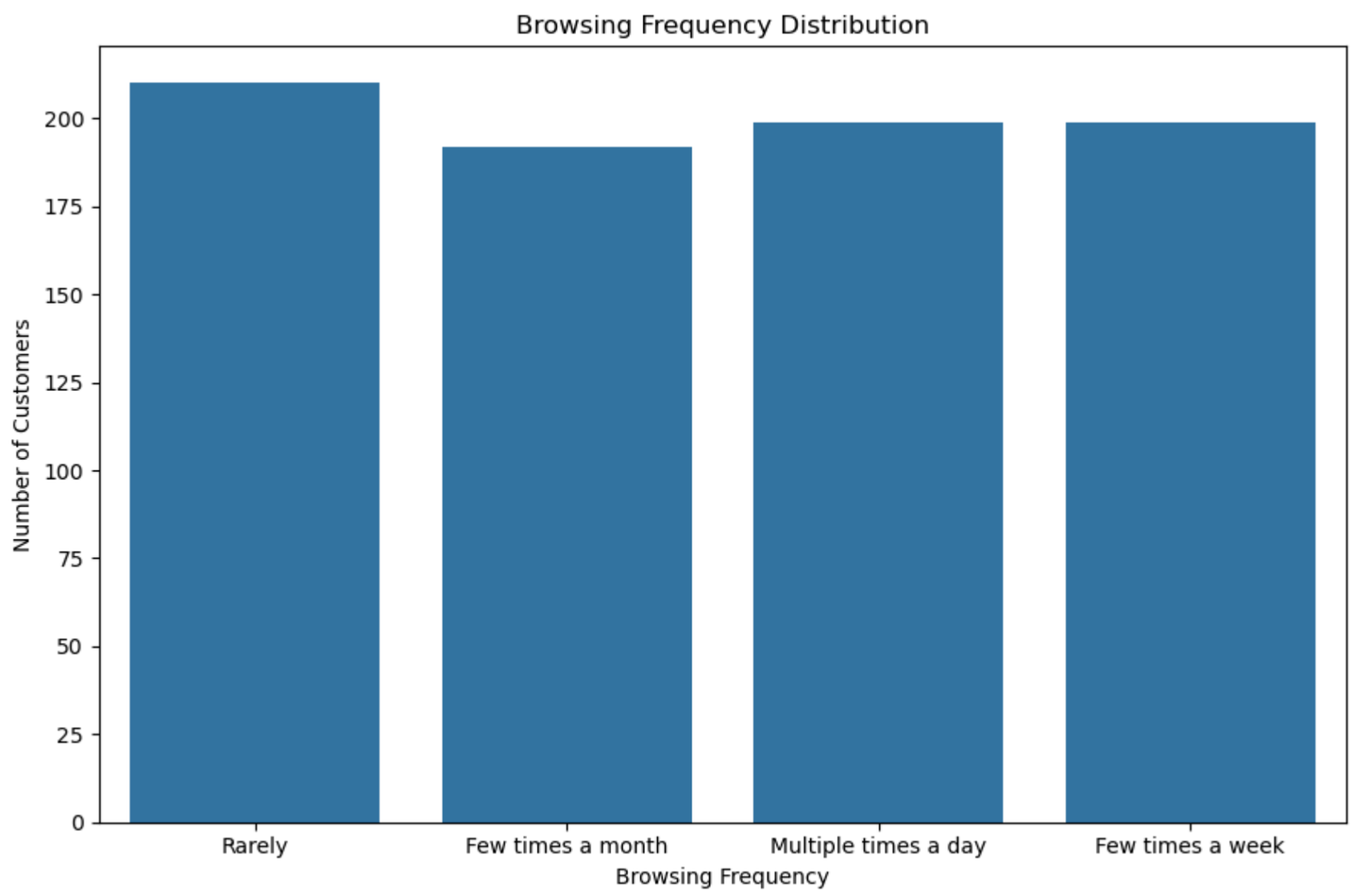
In [160…

```
## step 5.1: Create attractive visualizations (bar charts, heatmaps, pie charts) for:

## a:) Purchase categories

plt.figure(figsize=(12,8))
df['Purchase_Categories'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title("Product Category Distribution")
plt.ylabel("")
plt.show()
```



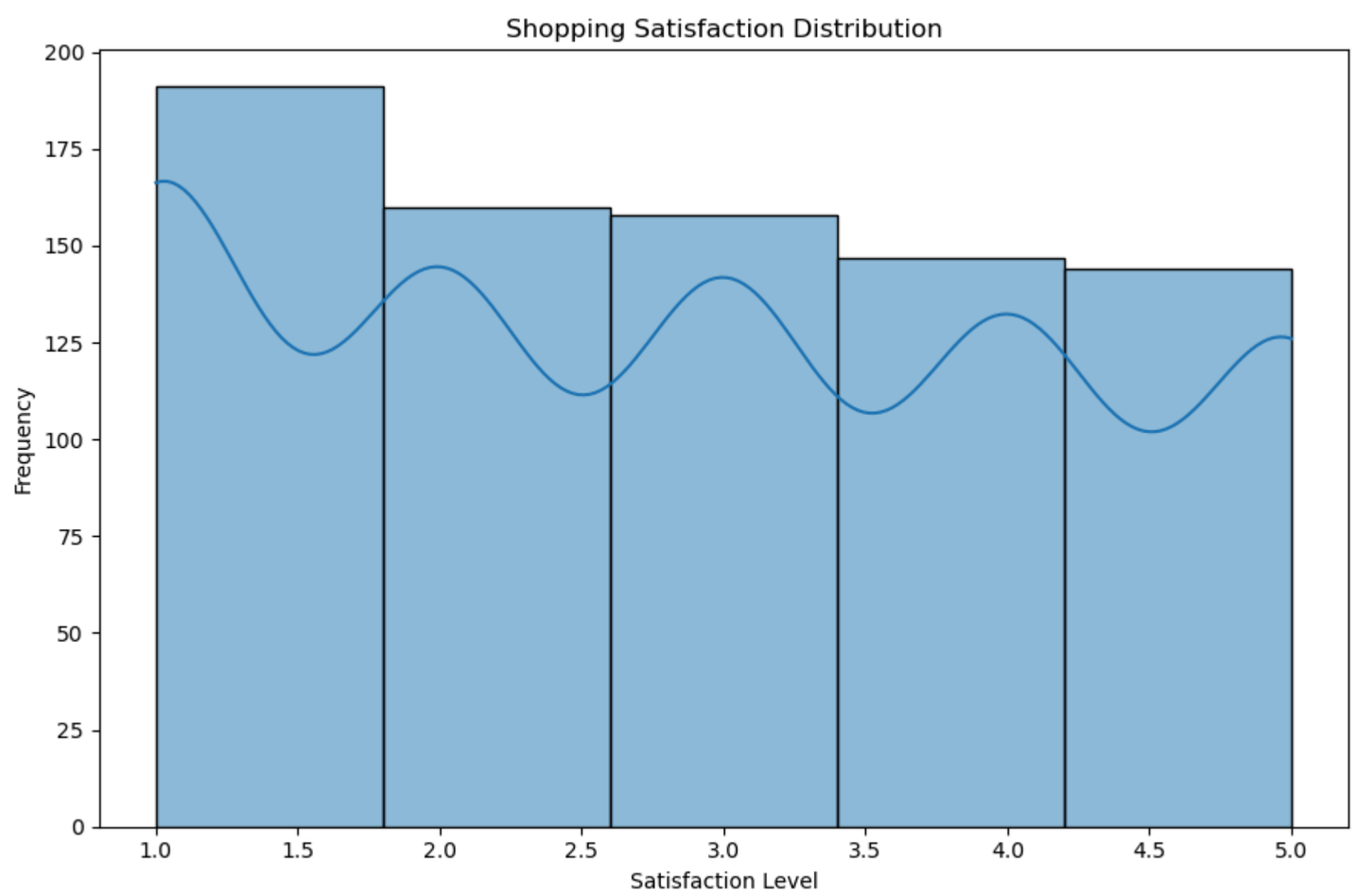
In [153…

```
## b:) Browsing frequency distribution

plt.figure(figsize=(9,6))

sns.countplot(
    x=df['Browsing_Frequency'],
)

plt.title("Browsing Frequency Distribution")
plt.xlabel("Browsing Frequency")
plt.ylabel("Number of Customers")
plt.tight_layout()
plt.show()
```



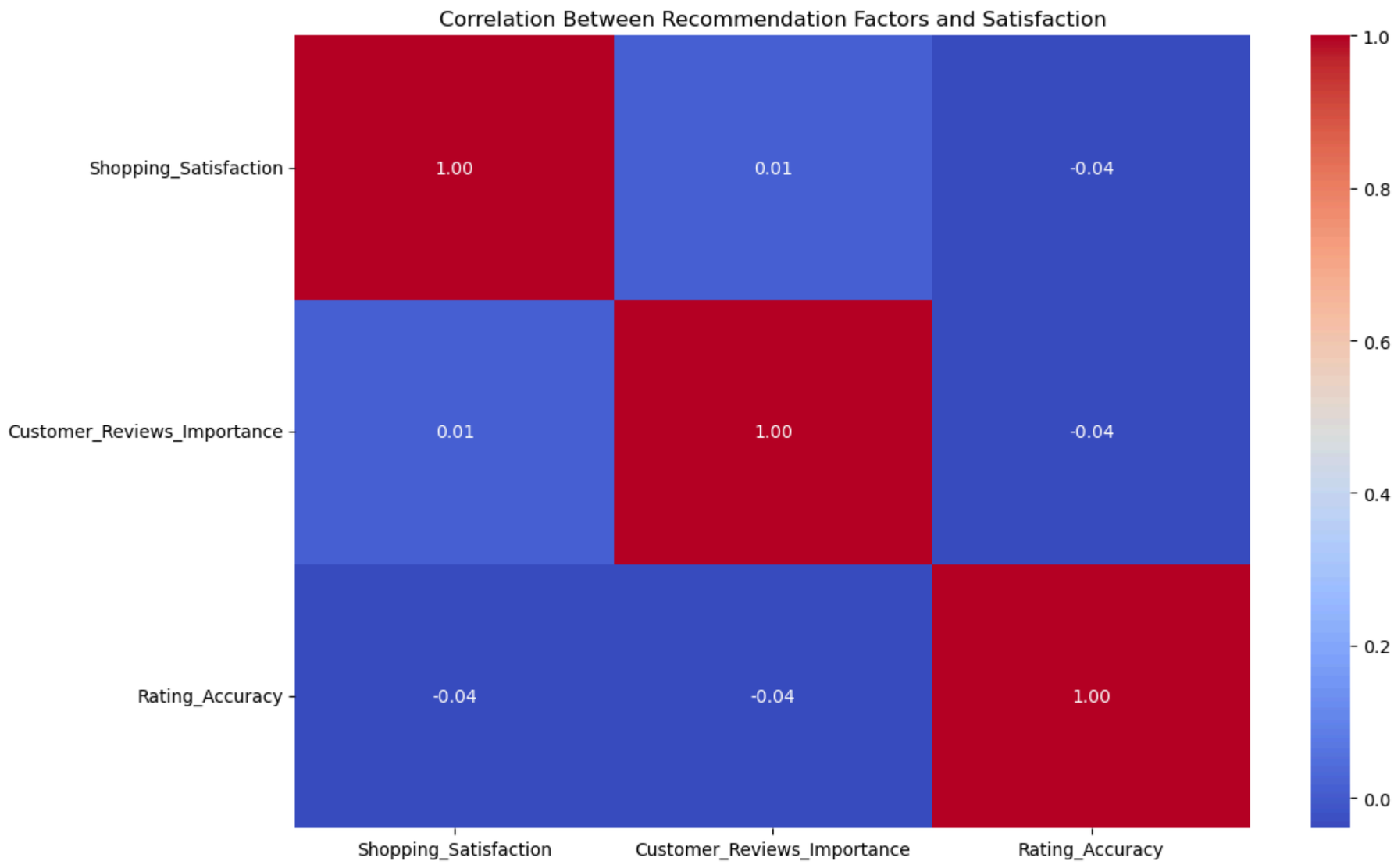
In [154…

```
## c:) Recommendation vs Satisfaction

plt.figure(figsize=(9,6))

sns.histplot(
    df['Shopping_Satisfaction'],
    bins=5,
    kde=True
)

plt.title("Shopping Satisfaction Distribution")
plt.xlabel("Satisfaction Level")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```



In [150…

```
## d:) dCorrelation: Recommendation Usefulness vs Satisfaction.

corr_data = df[
    ['Shopping_Satisfaction',
     'Customer_Reviews_Importance',
     'Rating_Accuracy']
].corr()

plt.figure(figsize=(12,7))

sns.heatmap(
    corr_data,
    annot=True,
    cmap='coolwarm',
    fmt=".2f"
)

plt.title("Correlation Between Recommendation Factors and Satisfaction")
plt.tight_layout()
plt.show()
```



STEP 2: Summary Findings (Use This in Report)

--> Key Insights

1. Purchase Categories A small number of product categories account for the majority of purchases. This indicates customer preference concentration, allowing better category-based recommendations.
2. Browsing Frequency Most users browse products frequently before making a purchase. This highlights the importance of personalized recommendations during browsing stages.
3. Satisfaction Levels Shopping satisfaction is moderately high for most customers. A smooth browsing and recommendation experience plays a crucial role in satisfaction.
4. Recommendation Usefulness & Satisfaction A positive correlation exists between recommendation usefulness and shopping satisfaction. Customers who find recommendations helpful tend to report higher satisfaction levels.

First video: ***e-bay presentation video no. I***

Second video:***e-bay presentation video no. II***

In [ ]: